

According to Manu, D.A.K., (2011), oil and gas revenues and royalties can positively impact countries of Africa, significantly delivering sustainable development gains to various communities including fisher communities. However, for this to happen, governments of Africa have to enact inclusive oil and gas sector policies that respond to the concerns and issues that act against the sustainability of sectors affecting communities. Practical realities are that fisher communities are displaced from lucrative fishing areas and replaced by Seismic Vessels whose noise disrupts fish breeding grounds, reducing catches, killing fish and drastically reducing fisher community incomes (Quijano 2008). Africa oil has mainly benefited African privileged elites, oil companies, their shareholders and western industrialized countries (ACODE, 2006) but this can be changed for a more favorable outcome. This study ventures into finding out the negative effects of the oil and gas industry on the fishing communities in Uganda, Kaiso, Hoima District and proposes responsive actions focused on responses that protect and promote the livelihoods of fisher communities.

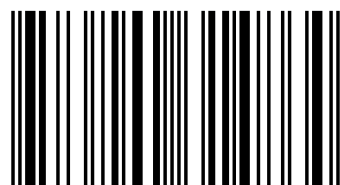


Edward Ssenyange

Effects of the Oil and Gas sector on Fisher Communities in Africa



Edward has 22 years of Development work with NGOs in Liberia, Sierra Leone, Uganda, and Tanzania. This includes private sector development work in poor countries to reduce poverty, strengthen economies, increase accountability and promote service delivery. He is currently spearheading a campaign for Africa to progress on SDGs achievement.



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**EFFECTS OF THE OIL AND GAS SECTOR ON FISHER COMMUNITIES
IN AFRICA: THE CASE FOR UGANDA, HOIMA DISTRICT, KAISO
LANDING SITE**

ENSURING SUSTAINABLE DEVELOPMENT FOR AFRICA

**By Dr. Edward Ssenyange
(PhD Sustainable Development)**

September 2019

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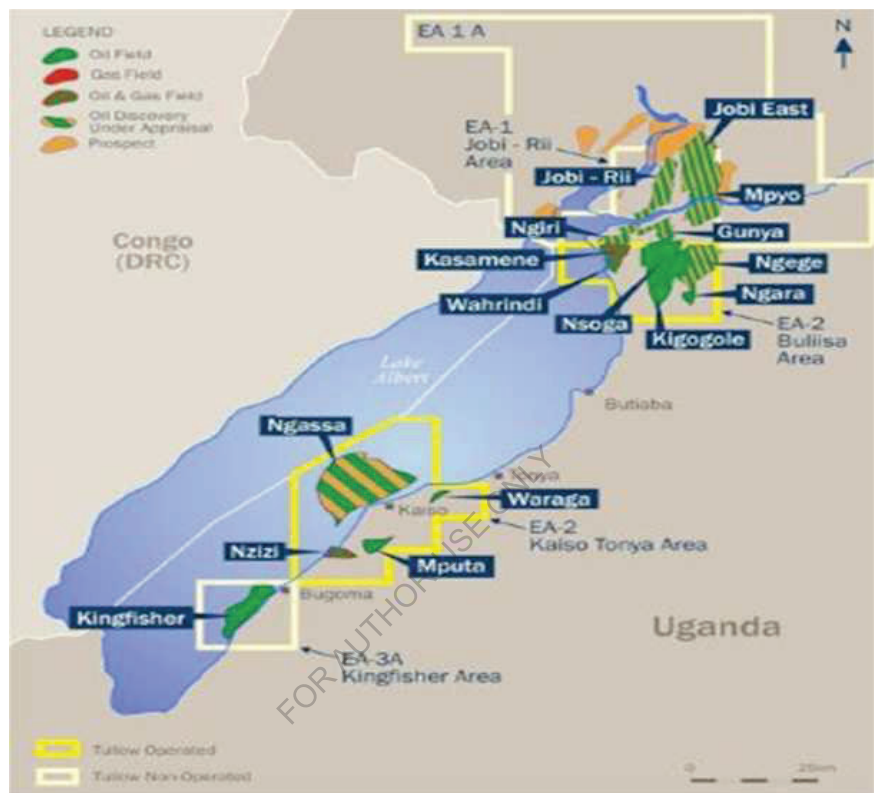
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ABSTRACT

According to Manu, D.A.K., (2011), oil and gas revenues and royalties can positively impact countries of Africa, providing significant sustainable development gains to various communities including fisher communities. However, for this to happen, governments of Africa have to enact inclusive oil and gas sector policies that respond to the concerns and issues that act against the sustainability of sector affected communities. Practical realities are that fisher communities are displaced from lucrative fishing areas and replaced by Seismic Vessels whose noise effect disrupts fish breeding grounds, reducing catches, killing fish and drastically reducing fisher community incomes (Quijano 2008). Africa oil has mainly benefited African privileged elites, oil companies and their shareholders and western industrialized countries (ACODE, 2006) but this can be changed for a more favorable outcome. This study ventures into finding out the negative effects the oil and gas industry on the fishing communities in Uganda, Kaiso, Hoima District and proposes responsive actions focused on responses that protect and promote the livelihoods of fisher communities. The study sets to establish relationships between the Independent variables which comprise Oil and gas exploration activities and the dependent variables that comprise factors constituting fisher community livelihoods. A review of literature exposed variables for the analysis which were useful in the building of the conceptual framework. The sample population constituted 120 persons with a sample size of 115 respondents ensuring confidence level of 95% with the margin of error at 1.96. The sample size was drawn from a cross section of Oil and Gas sector stakeholders including members of fisher communities in the targeted geographical area. Data analysis comprised both quantitative and qualitative methods with Relations between variables determined through correlation analysis with the use of Statistics Package for Social Scientists (SPSS). Conclusions expose acute gaps found by the study and address responsive actions comprising protection of fish habitats; management of pollution management on Lake Albert; Compensation, Capacity Building, Education and Restoration of livelihood frameworks for fisher communities; and enhanced stakeholder involvement and engagement in decision making processes. The adherence of such proposed actions by the relevant sector Ministries Departments and Agencies' will witness a milestone towards the achievement of inclusive oil and gas sector growth replicable throughout Africa.

Map 1: Location of Kaiso Landing Site on Lake Albert in Mid-West Uganda



SOURCE: www.mapof Oil Fields in Uganda

LIST OF ACRONYMS

ACODE	Action Coalition for Development
ALBERTINE GRABEN	Oil rich Rift Valley Region located around the Lake Albert which lies on the Uganda Western border with DRC
DFID	Department for International Development
DRC	Democratic Republic of Congo
EPA	Environmental Protection Agency
FPSO	Floating, Production, Storage and Offloading vessel
GDP	Gross Domestic Product
GNPC	Ghana National Petroleum Company
GoU	Government of Uganda
HC	Human Capital
HIV/ AIDS	Human Immunodeficiency Virus infection/ Acquired Immune Deficiency Syndrome
IRS	Internal Revenue Service
MFPED	Ministry of Finance Planning and Economic Development
MGLSD	Ministry of Gender, Labour and Social Development
PLC	Physical Livelihood Capital
PoD	Plan of Development
NLC	Natural Livelihood Capital
SC	Social Capital
SFLP	Sustainable Fisheries Livelihood Programme
SLF	Sustainable Livelihood Framework
TEN	Tweneboa-Enyenra-Ntomme
UPEPD	Uganda Petroleum Exploration and Production Department

CHAPTER ONE

INTRODUCTION

1.1 Introduction

This chapter presents the: background to the study; problem that has prompted the research; objectives of the study; research questions; scope in terms of time, technical and geographical scope; significance of the study; conceptual framework; and operationalization of the research.

1.1.1 Background

Fishing plays an important role in the socio-economic development of Uganda with fisher communities sourcing fish from Aquaculture and Natural water bodies specifically, rivers and lakes. The open water covers 15.3 percent of Uganda's surface and comprises of five major lakes namely Lake Victoria, Lake Albert, Lake Kyoga, Lake Edward and Lake George and these are the main sources of fish in the country. Lake Victoria continues to be the most important water body in Uganda both in size and contribution to the total fish sector catch, followed by Lake Albert and Lake Kyoga. Lake Albert/Mobutu lies along the Democratic Republic of Congo/Zaire-Uganda border in 43/57 per cent ratio in the faulted rift valley depression tending south-west to the north east. It is bounded by latitudes 1°0' N to 2°20' N and longitudes 30° 20' to 31° 20' E.

1.1.2 Global context

According to Manu, D.A.K., (2011), oil and gas revenues and royalties can provide significant support to the local economies of fishing communities. However, for this to happen, governments have to enact relevant laws and policies, ensure efficient implementation as well as monitoring. In the absence of appropriate policies, the local fishing economies will be adversely affected. The effects of Oil and gas industry on the fishing communities are that Fishermen are displaced from their lucrative fishing areas together with their boats and gears so as to provide the necessary space. Fishermen are replaced by Seismic Vessels whose first effect is to generate noise that disrupts fish breeding grounds. Seismic survey equipment used in offshore petroleum produces very loud noise and these explosives can kill fish in a range of hundreds of meters as well as reducing fish catches drastically (Quijano 2008). On sensing this activity, fish migrate several kilometers away from the air gun.

1.1.3 African Context

In Africa oil has mainly benefited African privileged elites, oil companies and their shareholders and western industrialized countries (ACODE, 2006). Since 1970, Nigeria's socio-economic and political fortunes have been intricately linked with oil exploration, with petroleum oil providing about 95 per cent of export earnings and accounting for over 80 per cent of government revenue as well as generating over 40 per cent of GDP. The Oil that generates these numerous benefits to Nigeria comes solely from the Niger Delta but has resulted into negative impacts to the communities and inducing a multifaceted problem in the region such as abject poverty, deprivation, social conflict, occupational dislocation, ill health (Amadi and Tomuno, 2012).

In the case of Ghana's oil and gas exploration, production is carried out offshore. The Jubilee Field is an oil field located 60 km off the coast between the Deep water Tano and West Cape Three Points blocks (Sakyi et al., 2012). According to Edjekumhene et al. (2010), the exploration and production of oil and gas may likely promote economic, social, environmental and cultural changes within the various fishing communities. This change could be either positive or negative but there is limited research on the impacts of the oil and gas industry on men and women working in the fisheries since commencement in the production of oil.

1.1.4 Ugandan context

Uganda discovered commercially viable oil deposits in the Albertine Graben in 2006 and has since embarked on establishing legal and regulatory frameworks for the sector to ably promote growth and development for the country. According to Ugandan Petroleum Exploration and Production Department report (2013), Uganda's proven oil reserves were estimated to be 3.5 billion barrels, which are expected to yield at least \$2 billion per year for 30 years once oil production commences. In light of these potentially transformative discoveries, Uganda faces a number of policy choices, and the welfare of the local communities including the fishermen and women in the Albertine Graben. According to Ministry of Finance Planning and Economic Development (MFPED), (2008; 2009; 2010; 2011; and 2012) fisheries is an important sector contributing about 4% to Gross Domestic Product (GDP) annually. This emphasizes the importance of fishing sector and communities to the Uganda economy. According to MS Logistics (2014) oil and gas exploration in the Albertine, Kaiso Landing site has resulted into: demarcation of exclusion zones which are *no-go* areas to fishermen; fish deaths and migrations; unviability of fishing activities and loss of

incomes; noise pollution; marine accidents; destruction of fishing gears by oil vessels; disruption of rainy seasons; and social networks. In other words, livelihoods social capital, physical capital, financial capital, natural capital, and human Capital were tampered with.

1.2 Problem Statement

According to Badgely, (2011), the oil and gas industry can impact negatively on fishing communities, An example is the case of coastal communities in the Niger Delta in Nigeria who have “suffered worst records of environmental pollution, violent deaths and chronic illnesses from regular pipeline and other explosions, oil rains, oil spills, oil fires, and water pollution” (des Clers, 2007: 31). Since the discovery and exploration of oil and gas in Uganda is quite new, the country is likely to suffer from potential risks and conflicts and consequential losses in the livelihoods of fishing families and communities in the Albertine Graben if it is not well managed.

1.3 Purposes of the Study

The study ventures into finding out the negative effects oil and gas industry on the fishing communities in Kaiso, Hoima District. Importance of findings will be of great potential benefit to solving potential explosive situations which could otherwise develop into insurmountable full-blown civil unrest.

1.4. General Research Objectives

To assess the effects of oil and gas industry on fisher community livelihoods at Kaiso landing site

1.4.1. Specific objectives

- i. To examine the effects of oil and gas surveying activities on changes in prices of commodities at Kaiso landing site
- ii. To determine the correlation between oil and gas exploration activities and community livelihoods
- iii. To examine road construction effects on changes in income in Kaiso Landing site, Hoima District

1.5 Research questions

- i. What are the effects of oil and gas exploration on changes in prices of commodities at Kaiso landing site?

- ii. What is the correlation between oil and gas exploration activities and the community livelihoods?
- iii. What are the effects of road construction on changes in income in Kaiso Landing site, Hoima District?

1.6 Hypotheses of the study

- i. Oil surveying operations have distorted price levels of basic commodities
- ii. There is a negative and positive correlation between oil and gas exploration activities and community livelihoods
- iii. Road construction activities have had positive effects on incomes but with some negative social impacts

1.7 Geographical, Subject and Time Scope

1.7.1 Subject/Content Scope

This study is concerned with finding out the effects of oil and gas prospecting on community livelihoods. It also interests itself in how much government sustainable livelihoods approaches have been sufficient in dissolving the negative effects.

1.7.2 Geographical Scope

The Geographical scope is Kaiso landing site on Lake Albert, Hoima District located in the western border regions between Uganda and DRC. The lake lies in the great East Africa Rift Valley.

1.7.3 Time Scope

The study examines effects during the period running from 2006 to 2016 – a ten (10) year stretch.

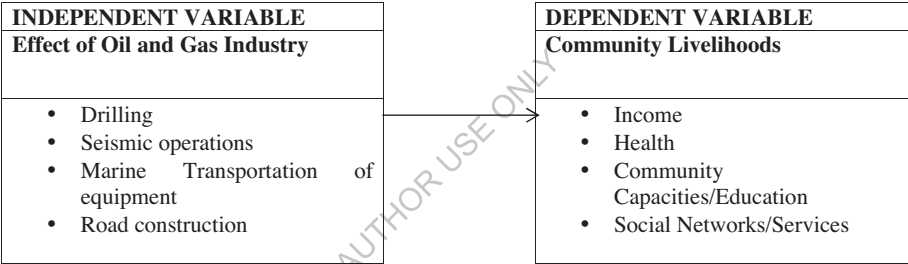
1.8 Significance of the study.

Study results are viewed as useful to regulatory bodies, parliamentarians, sector ministries departments and agencies, managers of oil companies, academia, and sector advocates like civil society organizations. Findings hold global stakeholders responsible for diffusing potential civil unrest circumstances let alone following up on all aspects of oil and gas related economic growth in the spirit of collective responsibility. It is useful to parliamentarians because it augments backup information to formulating necessary and sufficient policy. Sector ministries departments and agencies have gained information on what to enshrine in sector bills that they need to table on citizens and local company participation. Proprietors and managements of Oil companies are enlightened on social responsibility gaps. Academia gains extended areas for further research

while sector civil society groups gain action points for sector evidence-based advocacy and monitoring activities.

1.9 Conceptual Framework

The conceptual framework represents constituents of independent variables working upon the dependent variable, community livelihoods. Oil and gas exploration activities were analyzed as a cause of vulnerabilities to fishing communities’ in terms of **shocks, seasonality and critical trends**. Shocks include physical exclusion of fishermen from fishing grounds and drastic drops in catches; critical trends include oil spills and damping of poisonous waste; and seasonality refers to factors like changes in the breeding behavior and locations of fish as well as the impact of air pollution on weather, given that the rainy season is prime for fishing.



1.10 Operationalization of the Research

The research was implemented through a work plan found useful in observing a systematic approach. By and large, it was guided by a research problem, objectives, questions, and conceptual framework. It was implemented through the use of questionnaires; an interview guide and secondary data was used to contextualize the concepts and variables for the research as well as realize related case scenarios. A target population was identified, from which a sample was drawn. A pilot test was undertaken before the major launch of the main research to help check on the feasibility of the data collection tools.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews literature on the Sustainable Livelihood Framework (SLF); Sustainable Livelihood Assets; Oil and Gas Prospecting and Livelihoods of Fishing Communities in Ghana including remedial strategies and actions; Oil and Gas Prospecting and Fishing Community Livelihoods in Uganda; and the Research Gaps.

2.2 Oil and Gas Industry, Community Livelihoods and the Fishing Business

According to Chambers and Conway (1992), livelihood refers to capabilities, assets (stores, resources, claims and access) and activities required for a means of living. Livelihood are sustainable if they can cope with and recover from shocks, maintain or enhance capabilities and assets, create opportunities for future generations, and contribute net benefits to other livelihoods at various socioeconomic levels. The approach of the livelihood's framework is assessing the key causes of vulnerabilities. It takes into consideration the assets owned, controlled, and claimed by households (Allison and Ellis, 2001). Assets include incomes, health, knowledge and skills, friends and family (Rakodi, 2002). Sustainable Livelihood Approach is people-centered and seeks to identify and tackle the most pressing community constraints. It also considers the existence of capabilities of people to cope in times of intense danger such as droughts, climatic and seasonal changes. So for this study, the ability of fishermen and fish traders to use their assets such as canoes and fishing equipment to maintain their productivity in the advent of any major disturbance is key to resilience (Allison and Ellis, 2001).

2.2.1. Income

It is discussed under financial, physical, and natural livelihood capital dimensions

2.2.1.1 Financial livelihood Capital

According to (DFID, 2000), financial capital is a money resource and be converted easily into other forms of capital for direct achievement of livelihood outcomes i.e. purchasing of canoes for fishing or purchasing fish to process and sell. Financial capital can also be in a form of savings, credit, wages, proceeds from work and living costs. The production of oil and gas can lead to an influx of foreigners leading to a significant increase in the prices of goods and services. This adversely affects the cost of living of the fisher communities because they will rendered unable to

afford goods and services including rent that they were previously able to afford (Waskow and Welch, 2005). Offshore production activities threaten fish stocks resulting into sharp declines in fish yields which cripples household incomes of fishing communities.

2.2.1.2 Physical Livelihood Capital

According to (Boohene and Peprah, 2011) and (Rud and Aragon, 2013), physical capital comprises infrastructure and producer goods needed to support livelihoods, such as fuel, fishing nets, canoes, smoking ovens, secure homes and buildings, adequate water supply and sanitation, clean, affordable energy i.e. firewood for fish traders, and access to information. Oil and gas activities attract influx of opportunity-seeking populations as well as investments in social services. Associated with this influx is the cost of services including rents and rates for rental space. This can be stressful to fishing communities whose household incomes do not improve in accordance to other oil related trends.

2.2.1.3 Natural Livelihood Capital

According to Kollmair and Gamper (2012) natural capital is the natural resource stocks such as land, water, forests, air quality, erosion protection, biodiversity degree and rate of change. According to (Serrat, 2008), natural capital includes fisheries and is vital for community livelihoods particularly those that solely depend on nature. The Sustainable Livelihood Framework explains that there is a close relationship between natural capital and vulnerability.

Oil rigs stationed in breeding grounds for fish and other aquatic life disrupt breeding patterns and affect populations (Waskow and Welch, 2005). Oil spillages occur during drilling, pipeline leaks, and transfer-procedure of oil to tankers. Aquatic pollution from oil leads to the death of aquatic life. Also, the bright lights associated with the oil rigs attract fish into 'no go' zones where fishermen are not allowed to fish (Badgley, 2011).

2.2.2. Health, Community Capacities/Education, Social Networks and Social Services

They are discussed under human and social livelihood capital dimensions

2.2.2.1 Human Capital

According to (Eneyew and Bekele, 2013) and DFID (2000), Sustainable Livelihoods Framework (SLF) considers Human capital to mean skills, knowledge, ability to labor and good health that enable communities to pursue livelihoods of their choice. According to Allison and Ellis (2001), human capital is the most important asset of fishing communities and is enhanced by investments

in training, education, experience, and health (Ellis, 2000). Death, birth and migration also affect human capital.

With the emergence of oil and gas prospecting communities are expected to gain from lucrative employment opportunities but according to (Basedau, 2005), the oil sector employs mainly high-skilled workers, with limited opportunities for unskilled workers. This does not favor resilience for fishing communities because they are critically limited in skills (Waskow and Welch, 2005).

2.2.2.2 Social Capital

According to (Allison and Ellis, 2001), social capital is about community networks and is the stock of resources that may be actual or virtual and accrue to individuals and the community. Social capital can be durable and institutionalized and can be useful for boosting economic activity. It is also helpful for mutual acquaintance and recognition. Social capital emphasizes the importance of social interactions and structures to individuals and households. This capital is a social resource able to impact livelihood outcomes positively and can be networks and connectedness, that increase people's trust and ability to cooperate or membership in more formalized groups and their systems of rules, norms and sanctions. It also serves as security networks when fisher folk experience shocks. Oil and gas activities result in in-migration or out-migration which surely distorts social networks, damaging social security and resilience of fishing communities.

2.3 Oil and Gas Industry, Community Livelihoods and the Fishing Business in Ghana

According to European Conference on African Studies (2013) Ghana discovered commercial oil reserves in 2007 in the Deep Tano/West and Tweneboa-Enyenra-Ntomme (TEN) in 2010. According to Tullow (2010 the Plan of Development (PoD) of the TEN Project required the drilling and completion of up to about 24 development wells connected through subsea infrastructure to a Floating, Production, Storage and Offloading vessel (FPSO), moored in about 1,500 meters of water. The Owo field is estimated to hold between 70 and 550 million barrels of light sweet and high-quality crude respectively. Ghana has 550 km coastline and a total continental shelf area of about 24,300 square km that supports her marine fishing industry (Gyesi, 2012).

The industry provides a major source of inexpensive protein and many coastal communities depend on fisheries resources for their livelihoods. The fish sector is a major source of employment for coastal communities and contributes significantly to national income. With the introduction of off-shore oil and gas activities serious inter-sectors challenges have emerged. These emanate from

an intersection of sector interests arising from Lucrative fishing grounds being demarcated as no go areas to fishermen since they are also prospecting sites. This is hazardous to the fish sector because, according to the Sustainable Fisheries Livelihood Programme (SFLP) in Ghana, the fisheries sector triggers beneficial multiplier effects such that one fishing job has the effect of creating about seven others. Created jobs include supply of goods and services that are inputs to the fish sector production process i.e. boat-building, fishing net production, supply of fuel, marketing, processing and transport services. So, the loss or threat to one fishing job does not affect a single community but has negative multiplier effects across sectors and is a pointer to the fact that the fishing economic activity deserves critical attention.

According to (CRC 2010) and (Scoones 2009) Sustainable Livelihoods Approach (SLA) was applied to the fisheries sector as an analytical tool to explore links between fisheries livelihoods and the larger context of oil and gas prospecting. The analysis confirmed that oil and gas activities were indeed a stress to the fisheries sector. The SLA found that capital assets owned and controlled by fishing communities are mainly skills, knowledge and the ability to engage in fishing as an economic activity which forms their human capital base. Communities possess networks that rely on relationships to thrive and these comprise social capital. In terms of physical capital, most fishermen rely on the basic materials, tools and equipment which can be considered elementary. The financial capital base comprises of access to savings and credits. However, for the oil and gas sector, human capital is driven by dependence on high level technical skills; alongside the need for an extremely sophisticated technology which is mostly very capital intensive. Unfortunately, these technical skills are not available to local communities and Livelihood sustainability policy has to address such issues. In this case, social capital includes the power to negotiate for new rules to influence the patterns of government policies and the extent to which these policies and rules can be adopted and enforced in the interest of fisher-communities or oil and gas companies. It is important for sectors arrangements by government and stakeholders to develop local expertise to take advantage of the benefits from the oil and gas sector. Policy measures can provide for expatriate quotas in oil and gas prospecting to allow opportunities for local content. Communities can also gain from policy that provides for a Public-Private-Partnership between government and oil companies to provide basic infrastructure and amenities such as better roads, public toilets, boreholes, schools and health facilities. Scholarships schemes could also be put in place to support youth education. Such efforts ease social tensions and are key in averting strife.

(CRC 2010) acknowledges a decline in traditional fishing grounds and catching opportunities for fishermen due to increased competition from displaced fishing grounds. In the event of establishment of exclusion zones, fishermen have to circumvent new routes to access new grounds, taking more travel time. Since the two sectors of oil and gas and fishing are inter-twined, it is important to forge workable co-existence and thrive. A workable approach would be to utilize part of the oil and gas revenues for promoting the fish sector, and integrating fishermen into co-existence of the two sectors. However, without conscious efforts towards this direction, it is unlikely that negative impacts on fishing communities and the fish sector will be resolved.

Actions directed at integrating the two sectors can be incorporated into comprehensive awareness plans that will target challenges like the presence of fishing boats in areas where the oil and gas platforms are located i.e. in the navigation exclusion zones while paying attention to protection of Livelihood Assets. However, there is a need for stakeholders to carry out monitoring and evidence-based dialogue. CRC also asserts the need for the establishment of a Commission for Oil Sector Impact Assessment on the fisheries and fishing communities.

According to Egyir Kwasi Isaac (2012) government in Ghana should ensure Good environmental management and oil spill response. A natural resource fund is Proposed to cater for compensations and environmental management. Fishermen need to be assured that fishing communities will be compensated in case of oil spills and livelihood distortions. Also proposed is intensive oil monitoring activities in order to reduce oil spills and other adverse impact from the oil and gas activities that might affect the fish stocks and marine ecology. A successful co-existence of the sectors is vital to avoid the Dutch Disease which takes root when the oil and gas sector cripples' other economic sectors of the oil resourced economy.

Egyir Kwasi Isaac (2012) asserts that it is important to manage service delivery in terms of amenities like housing, schools, and health facilities due to the migrations that follow oil discoveries as people flock into the zones to take advantage of opportunities. If not properly managed, social pressures that degenerate into conflicts can emerge and distort economic gains. Another feature of the oil and gas sector that has to be addressed is the skyrocketing of prices and inequalities in social classes, where the rich continue to be rich and the poor get poorer. Local fishermen livelihoods and participation can be improved and social pressures reduced through

inter-sectors activities like integrated security and performance monitoring, control and surveillance of the coastal regions.

There is also need to improve stakeholder involvement and institutional capacities towards public-private partnership, frequent consultation and dialogue with the people, traditional leaders, civil society organizations and countrywide education using the religious bodies, the media and various traditional management and authority that govern fishing operations in the regions. Institutions such as the Environmental Protection Agency (EPA), the Internal Revenue Service (IRS), the Navy, the Ghana Maritime Authority, the Fisheries Commission, the Banking and Insurance industries need to be well resourced with personnel and equipment needed to enable them function properly. Good governance and accountability are pivotal in Ghana's petroleum and revenue management but should not be left to one body i.e. GNPC which is only responsible to the president and not Parliament. It is proposed that no secrecy should exist in oil transactions between foreign oil companies and GNPC.

2.4 Oil and Gas Industry, Community Livelihoods and the Fishing Business in Uganda

Uganda discovered commercially viable oil deposits in the Albertine Graben in 2006 and has since embarked on establishing effective resource management procedures to promote growth and development. By the end of 2013, Uganda's proven oil reserves were estimated by the Uganda Petroleum Exploration and Production Department (UPEPD) to be 3.5 billion barrels. It is anticipated that once oil production commences, the reserves will yield at least \$2 billion per year for a period of 30 years. Bank of Uganda estimates that the country could save up to US\$630 million annually on oil imports once production commences. In light of these potentially transformative discoveries, Uganda has the fundamental responsibility of balancing economic gains and socioeconomic welfare of local communities.

This study interests itself with the effect of oil and gas activities on livelihoods of fishing communities in the Albertine Graben's Kaiso Landing site. According to (Darkwa, 2010), the discovery of oil and gas in Uganda sparked high social and economic expectations. Following the consequential offshore oil and gas activities in Uganda public debates have emerged expressing deep concerns about the impact on the local community's wellbeing. The study focuses on fishing communities because the fishing sector plays a key role in exports generation for Uganda as well as household incomes for the fishing communities. Besides this matter is the fact that considering

the financial yields expected from the oil and gas sector, it should be seen to not only integrate well with all other economic sectors but to support sector wide sustainable development rather than retrogression.

Uganda sources its fish from aquaculture, rivers and lakes. Rivers and lakes cover 15.3% of surface land with the major lakes being 5 namely: Victoria, Albert, Kyoga, Edward and George. Lake Victoria is the most important water body in Uganda both in size and contribution to the total fish catch. It is followed by Lake Albert located in the Albertine Graben and lies along the national border between the Democratic Republic of Congo (DRC) and Uganda in 43/57 per cent ratio. The lake is a result of the formation of the East Africa rift valley millions of years ago and lies in a faulted depression trending south-west to the north east and bound by latitudes 1°0' N to 2°20' N and longitudes 30° 20' to 31° 20' E.

According to Ministry of Finance Planning and Economic Development (MFPED), (2016) the fish sector contributes about 4% to Gross Domestic Product (GDP) annually. Mills et al. (2004: 13), underscoring the great importance of fishing to community livelihoods in terms of household incomes and subsistence for communities like those in Kaiso on Lake Albert. Communities depend on fishing and fish related activities for their livelihoods.

According to MS Logistics (2014) the oil exploration phase significantly altered the fish habitats and the viability of fishing activities on Lake Albert. Fishing grounds faced restrictions and fishermen were prevented from fishing in oil exploration waters and never compensated for the loss of incomes and food. Fishermen lost fishing gear destroyed by prospecting vessels of sector companies. Large diesel-powered vessels scared away fish from breeding and feeding grounds, resulting into drastic decline in fish catches. In addition, gas flaring during oil exploration destabilized the regularity of rainfall which is often consistent fish populations since fish stocks tend to increase during rains.

The oil and gas industry has a large potential to positively impact the livelihoods of communities. The sector is able to provide jobs and revenue accrued from the oil and gas which could be used to support and strengthen the economies of fishing communities (Egyir, 2012). For example, investment in improved infrastructure, water supply, sewerage and waste treatment, health care and education would bring about improved community welfare (E&P/UNEP, 1997). According to (USAID 2014), some known Corporate Social Responsibilities (CSR) of oil and gas companies in

some fishing communities include support to health initiatives that address preventable diseases by strengthening health systems as well as generating awareness and understanding of health issues (Kirkbride and Kenlay, 2012). For instance, Tullow supported the construction of Kaiso Primary school in Hoima district to provide a better educational infrastructure to the community. The school has two classroom blocks, three staff houses, kitchen and lavatories. Tullow Oil has also constructed Buliisa Health Centre IV at approximately \$2m with intentions to improve health systems in Buliisa District. Tullow Oil upgraded the Kyehoro and Kaiso primary schools (in Kaiso) with new infrastructure such as class room block for 21 classes against one block, staff quarters, water and sanitation facilities, classroom furniture and textbooks for the pupils in 2012/2013. Tullow has also worked closely with the communities, local authorities and health partners in establishing one of the company's key infrastructure projects, the Kyehoro Health Centre, based in Kaiso near Lake Albert. The health Centre supports the communities with various health services including HIV/ AIDS prevention and treatment, immunization, malaria treatment, family planning and antenatal services to the fishing communities in Kaiso (Panford, 2010). Oil and gas sector activities supports allied services like hotel lodgings, restaurant services, transport activities, road construction, and telecommunication, (Boohene and Peprah, 2011). Some of these facilities may be accessible for use by fishermen and fish traders. All this infrastructural development and satellite businesses provide new fish markets with new and better roads aiding easier access markets.

According to (Badgely, 2011), despite promising local community developments gains expecting from oil and gas prospecting, seriously negative impacts on fishing communities are looming. This study is conscious of the fact that the oil sector in Uganda is in its infancy. Despite this it underscores the importance of strengthening the relevant sectors wide operating environment to ensure integrated positive impacts across the economy beginning with communities located in the oil and gas sector prospecting geographical regions. Warning bells for Africa oil rich economies is the case of the persistent social instabilities among the coastal communities of the Niger Delta in Nigeria that have suffered serious environmental pollution, violent deaths and chronic illnesses from regular pipeline and other explosions, oil rains, oil spills, oil fires, and water pollution" (des Clers, 2007: 31).

According to International Alert (2015), Oil exploitation in Uganda holds strong promises of economic growth able to usher the country into a middle-income country status, which means that

several communities could be saved from poverty. The above notwithstanding, experience in other mineral-rich economies, especially in Africa, assert that for this to happen, it is important to get all strategies and actions leading to the desired status correct. Current fears among stakeholders in Uganda are that GoU views oil prospects as a quick economic solution and is overlooking the dangerous underlying issues facing the country like inequality, corruption, land disputes, all of which are likely to be exacerbated by oil exploitation.

2.5 Oil and Gas Industry and Community Livelihoods Policy for Uganda

According to the Ministry of Gender, Labor and Social Development, (2015), Social protection refers to public and private interventions in place to address risks and vulnerabilities that expose individuals to income insecurity and social deprivation, leading to undignified lives. It is a basic service and a human right that ensures dignity of people. In the Ugandan context, the social protection system is comprised of two pillars, namely: social security; and social care and support services. Social security refers to protective and preventive interventions to mitigate factors that lead to income shocks that affect consumption.

Social protection provides a secure platform upon which individuals can build productive and sustainable livelihoods, directly reduces poverty and vulnerabilities, as well as supporting excluded citizens in accessing other services. It enables citizens to lead secure and dignified lives and empowers them to contribute to and benefit from economic growth.

Prudently, Governments all over the World recognize Social Protection as a crucial component of national development strategies for achieving inclusive, pro-poor and equitable growth. Social Protection has emerged as an important intervention for supporting the population to maintain a minimum acceptable level of consumption and live dignified lives. It strengthens the capacity of the vulnerable persons and their families to meet basic needs, build resilience, access opportunities and improve life chances.

2.6 Research Gaps.

The majority of studies in Uganda examine the comprehensiveness of Production Sharing Agreements; Environmental Impact Assessments; and Accountability Aspects of Government and Oil Companies transactions. There exist interview records of local communities by civil society

organizations but studies specific to Kaiso fishing communities are not analyzed in terms of the Sustainable Livelihoods Approaches.

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CHAPTER THREE

METHODOLOGY

3.1 Introduction

Below is a breakdown of the research approach which comprises the research design; variables for the analysis; targeted sample population; sample size; data collection tools; data source; study type and anticipated challenges.

3.2 Research Approaches

According to (A.P. Kelly, 2016), central to the research was a research problem which was investigated by a set of objectives and research questions. A review of literature helped to identify variables for the analysis which were utilized in building the conceptual framework. Primary data was collected using questionnaire and interviews method while secondary data was collected from works already accomplished by various scholars and acknowledged herein. Analysis of data utilized both quantitative and qualitative approaches. The qualitative approach was premised on observation, deductions and recording. This helped to corroborate target group respondents and stakeholder interpretations. Quantitative strategy was variable centered involving selection and counting of target categories and finding out relationships of the research variables, trends and associated explanations. Systematic randomly sampled data was used with accuracy of descriptions for the population estimated at 95%. Relations between variables were then determined through correlation of Multi-dimensional attitude scaling.

3.3 Research Design

Research design provided a framework for the collection and analysis of data. This framework constituted of three dimensions namely: the scale of data collection; the use of time order; and inter-group comparisons. Research design opted to study a sample as a representation of the larger affected population which provided for an in-depth study of a representative sample. The units of analysis were people belonging to the various stakeholder categories. A scale was used to determine the actual number of persons per category and a special context-based selection on proportions of stakeholder numbers, was used and helped to add to the concept of representativeness.

3.4 Sample Population

The study targets 120 persons who are Oil and Gas sector stakeholders comprising of: Officials from local government; Officials from ministry of fisheries and aquaculture; Chiefs of fish traders; Chiefs of fishermen; Fish folks; Canoe owners; Officials from oil companies; Naval officials; and Traffic officers in Kaiso, Hoima District.

3.5 Sample size

The sample size selected from a population of 120, is 115 respondents, with the margin of error at 1.96 and confidence level of 95%.

$$\text{Sample Size} = \frac{\frac{z^2 \times p(1-p)}{e^2}}{1 + \left(\frac{z^2 \times p(1-p)}{e^2 N} \right)}$$

Population Size = N | Margin of error = e | z-score = z (<https://www.surveymonkey.com/mp/sample-size-calculator/>)

(Levy and Lemeshow, 2013).

The formula was used to determine the sample size and below, reference is also made to the sample size estimation table

SAMPLE SIZE ESTIMATION TABLE

R.V. KREJCIE AND D. W. MORGAN (1970) SAMPLE SIZE ESTIMATION TABLE

N*	S [†]	N	S	N	S	N	S	N	S
10	10	100	80	280	162	800	260	2800	338
15	14	110	86	290	165	850	265	3000	341
20	19	120	92	300	169	900	269	3500	346
25	24	130	97	320	175	950	274	4000	351
30	28	140	103	340	181	1000	278	4500	354
35	32	150	108	360	186	1100	285	5000	357
40	36	160	113	380	191	1200	291	6000	361
45	40	170	118	400	196	1300	297	7000	364
50	44	180	123	420	201	1400	302	8000	367
55	48	190	127	440	205	1500	306	9000	368
60	52	200	132	460	210	1600	310	10000	370
65	56	210	136	480	214	1700	313	15000	375
70	59	220	140	500	217	1800	317	20000	377
75	63	230	144	550	226	1900	320	30000	379
80	66	240	148	600	234	2000	322	40000	380
85	70	250	152	650	242	2200	327	50000	381
90	73	260	155	700	248	2400	331	75000	382
95	76	270	159	750	254	2600	335	100000	384

*N is the population, [†]S is the sample size

3.6 Sample Selection

According to Rice (2010), sampling entails gathering “information about a relatively small part of a larger group or population in order to make inferential generalization about the larger group. Elements of the sample were selected using two sampling techniques which are, purposive and snowball sampling methods. Researchers using the purposive sampling technique handpick the informants to be included in the sample on the basis of their judgment of typicality or possession of particular characteristics being sought (Cohen, 2007). As its name implies, samples are selected for specific purposes. Usually, it is used in order to access ‘knowledgeable people’, i.e. those who

have in-depth knowledge about particular issues, by virtue of their professional role, power, access to networks, expertise or experience (Ball, 1990, cited in Cohen, 2007). The snowball sampling technique was used to get access to fish traders and fishermen representation. Snowball sampling involved selecting informants with ‘necessary characteristics’ who through their recommendations help the researcher in finding other informants (Seale et al., 2004: 449; Bryman, 2008).

Table 3.1: Sample Selection Content

No.	Strata	Number of respondents
1	Officials from local government	4
2	Officials from ministry of fisheries and aquaculture	4
3	Chiefs fish traders	16
4	Chiefs fishermen	16
5	Fish folks	23
6	Canoe owners	18
7	Officials from oil companies	16
8	Naval officials	16
9	Traffic officers in Kaiso	2
	Total	115

Sources: Primary data 2018

3.7 Data Sources and Collection Methods

Cooper and Schindler (2006) states that data refers to facts and perceptions emanating from real-life experiences collected from respondents. Data for the study was both primary and secondary. Secondary data was sourced from Analysis Reports by Academia and Sector Civil Society Advocacy Documents. Primary data collection considered the various stratum as shown in table 3.1 above.

3.8 Data Collection Tools

The research utilized questionnaires and interview guides for data collection. These tools were well responded to except for some delays experienced subject to the very busy schedules of some respondents. This delayed the return of filled questionnaires from respondents.

3.9 Research Procedures

A pilot survey was launched before the major research in order to ascertain that the data collection tool was appropriate before coming up with the final questionnaire. According to (Cunningham

2012), a pilot test is finding out if key informant targeted by a survey interview guide or observation form will be effective in generating the required information by first trying it out on a few people. Through this, the clarity of questions and precision are evaluated (Levy and Lemeshow, 2013) and in case there is need, amendments are made before launching the full-scale research. In addition, the pilot test also reveals the validity of the respondents.

3.10 Data Presentation, Analysis and Interpretation

Data was analyzed and presented through utilization of Ms EXCEL and SPSS to generate pie charts and correlation tables. These formed the basis for interpretations

3.11 Limitations and Mitigations

Some of the targeted respondents were persons engaged in off-shore activities and as search were not available on the mainland all the time and yet the researchers needed to follow-up on their responses. Mitigation was mainly an exercise of patience, follow-up calls and getting their respondent colleagues to remind them, which happened to work very well.

CHAPTER FOUR

PRESENTATION AND INTERPRETATION OF FINDINGS

4.1 Introduction

Following are: demographics of respondents i.e. Gender, Age, Marital Status, and Level of Education; findings on relations between independent variables i.e. Drilling, Seismic operations and Marine Transportation and dependent variable, Community Livelihoods i.e. Income changes; Health; Education, Electricity and other Social services. Included are findings from guided interviews with the sample population.

4.2 Demographic Data: Characteristics of Respondents

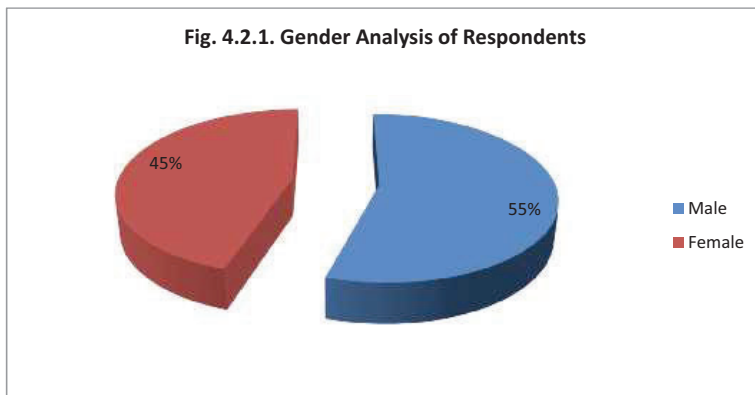
The respondent's bio data was analyzed in terms of gender, age distribution, level of education, marital status, and type of enterprise in order to determine their contribution relevance to the area of study.

4.2.1 Gender of the Respondents

Table 4.2.1 Gender Analysis of Respondents

Gender	Frequency	Percent	Cumulative Percent
Male	63.0	54.8	54.8
Female	52.0	45.2	100.0
Total	115	100.0	

Source: Primary data (2018)



Source: Primary data (2018)

Figure 4.1.1 depicts respondents’ genders as being 55% males and 45% males. It was found useful to collect data from both genders to aid in any analysis that may be necessary respective to gender and in anticipation of specific challenges faced by the different gender categories in the fishing community. The number of women participating in the sector is close to that of men because while men do the fishing, many women are engaged in the drying and selling of fish. However, other strata also contributed to the numbers.

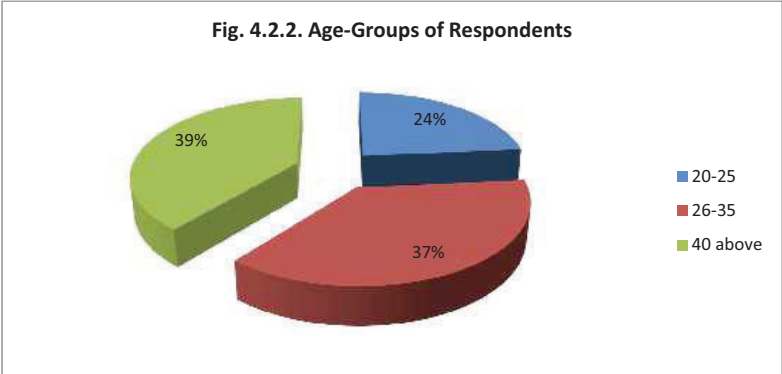
4.2.2 The Age group of the Respondents

It was also found useful to categorize respondents according to age groups in order to capture specific issues respective to age differences. Below is an analysis of age group data.

Table: 4.2.2 Age group of Respondents

Age Group	Frequency	Percent	Cumulative Percent
20-25	27.0	23.5	23.5
26-35	43.0	37.4	60.9
40 above	45.0	39.1	100
Total	115.0	100.0	

Source: Primary data (2018)



Source: Primary data (2018)

For all charts below, data was entered into MS excel sheets and used to generate representative pie charts so Figure 4.1.2 depicts respondent’s age groups as: 24% of respondents being between 20-25 years of age; 37% being between 26-35 and 39% being above 40 years. This information is imperative since the different age categories may have peculiar experiences to share, relevant to livelihoods challenges faced. It is noted that each of these categories have varying loads of societal responsibilities. However, other strata also contributed to the numbers.

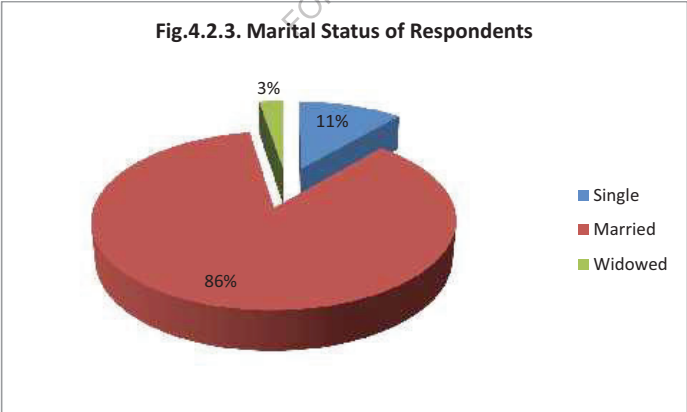
4.2.3 Marital Status of the Respondents.

Data for marital status is presented below and again, these categories are useful due to their peculiar exposures to responsibility. The researcher noticed that fishing communities assumed marriage at an early age and this contributed to the high percentage of married respondents.

Table: 4.2.3. Marital Status of Respondents

	Frequency	Percent	Cumulative Percent
Single	13	11.3	11.3
Married	99	86.1	97.4
Widowed	3	2.6	100
Total	115	100.0	

Source: Primary data (2017)



Source: Primary data (2018)

Figure 4.2.3 analyses respondents’ marital status as married being 86%, Singles being 11%, while widowed were 3%.

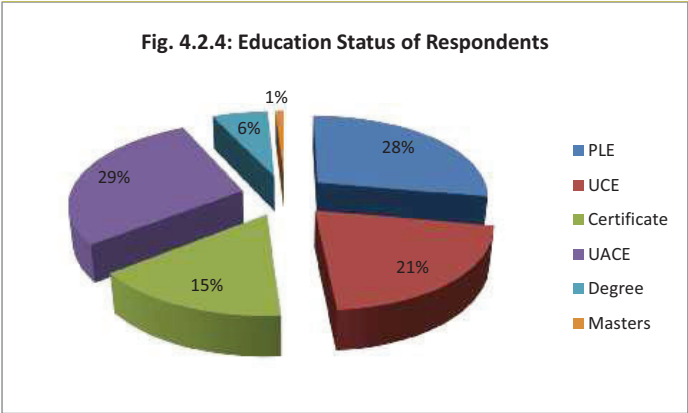
4.2.4 Education Level of Respondents

Analysis of education status was considered important because a basic education is necessary for persons to take advantage of most gainful opportunities available in the oil and gas sector. Also, persons with a basic education are a highly trainable resource in order to en-skill and orient them to any sector. Additionally, levels of education help in the internalization of policy provisions and articulation of sector issues.

Table: 4.2.4 Education Level of the Respondents.

Education level	Frequency	Percent	Cumulative Percent
PLE	32	27.8	27.8
UCE	24	20.9	48.7
Certificate	18	15.6	64.3
UACE	33	28.7	93.0
Degree	7	6.1	99.1
Masters	1	0.9	100
Total	115	100.0	

Source: Primary data (2018)



Source: Primary data (2018)

Figure 4.2.4 is a representation of the various categories of education of the respondents and shows that: 28% were Primary School Level Exams (PLE) certificate holders; 24% were holders of Uganda Certificate of Education (UCE); 11% were certificate holders in technical, teaching and other various skills; 6% were Bachelor's degree holders; 1% were holders of Master's Degrees. In general, respondents were found able to articulate sector issues even for community members who preferred to speak in their native language.

4.3 Results from Correlation Analysis of: Drilling on Community Livelihoods and Fishing Business; Seismic Operations on Community Livelihoods and Fishing Business; Marine Transportation of Equipment on Community Livelihoods and Fishing Business

Utilizing responses from questionnaires and interviews, data was entered into MS SPSS and correlations analyses was done as reflected in Appendix II. Using 2 tail test and bivariate analysis, all correlations were found significant at 0.01 levels. This confirmed that oil and gas industry affect livelihoods and the Fishing Business. This leads to the deduction that the management of government livelihoods approaches is of vital importance in checking the negative effects and social discontent that may arise.

4.4 Results from Deductive Analysis

Deductions were collected from questionnaires and interviews and corroborated with district local government reports and other stakeholders like International Alert. By and large, the oil sector has had positive and negative effects on the fishing communities of Kaiso. Positive impact are associated with social services infrastructure like improved access to fresh water for production of ice in fish processing; improvement of access roads in terms of networks and road quality; improved supply of fuel for motor boats; docking facilities; provision of electricity domestic, street lighting, icing and cold storage; improved communications between fishermen and markets and also for communication in the event of emergencies while fishing far offshore in marine and lake environments. Indeed, these positive effects to some extent impact positively on Human, Financial, Social, Natural and Physical capital but the net position of the impact is most important.

4.4.1 Oil and Gas Industry, Income changes, Health, Education and other Social Services

Negative impacts were identified on fish stocks, the environment, reduction in the area accessible to fisheries, increased risk to life and equipment, physical displacement of communities, increases in population and behavioral changes, and sharp decline in household incomes. These effects were

expressed as a serious challenge that carries a risk of undermining the oil sector as well as spreading civil unrest.

Each negative impact mentioned above expresses the effects that Oil prospecting has impacted on the physical environment and destroyed the biophysical resources that support fisheries, disrupting the fisheries habitat, and adversely affecting fish resources. In some cases, construction and operations activities have been associated with irreversible loss of fish habitat. Jetties and breakwaters also devastate fish habitats through their effect of disturbing water currents and the deposition of sands. Other destructive activity is construction of dams and operations that result in changes in downstream flow regime; and development of mines in watersheds. Development and installation of facilities on lake shores; use of surface water and groundwater; increased presence of water-impervious surfaces; earthworks that lead to the in-fill of near-shore environments, such as wetlands, and changes to shores; and dredging to create, expand, or reinforce channels for shipping are all destructive phenomena. Physical disruption of the Lake and its shores has led to the decline in stock, quality, and productivity of fish resources. Project activities may be associated with increased turbidity and acoustic disturbance, leading to alterations in the benthic habitat and negative effects on benthic fauna, fish avoidance, and fish mortality.

The disposal and discharge of project waste products negatively affects fish habitat by creating adverse conditions for fish. Infrastructural developments like dams, roads and other utilities that cross streams and near-shore infrastructure (e.g., piers) impair fish migration, as well as associated spawning. In addition to the physical structure itself, noise, vibration, toxic spills, and increased turbidity associated with construction can disrupt migration and spawning behavior and may cause acute or latent mortality of eggs, larvae, and juvenile stages of many species. New piers, even if on pilings, impact small or juvenile fish migrating along the shore. Several species of fish characteristically do not move through dark shadows cast by structures like overwater piers and in detouring course, are forced to move into deeper waters. In deep waters, fish are exposed to predators. Also, predatory species can hide under piers or around pilings to ambush small migrating species. Tampering with Natural Livelihood Capital has caused direct implications on household incomes hence Financial Livelihood Capital.

Oil and gas prospecting increased risks to fishermen and their equipment with most significant associated with increased number of vessels, their movement, and the impacts associated with their

movement i.e. waves from outboard motors and the large size of bow waves they create, as well as accidental collisions. Establishment and enforcement of exclusion zones increases safety risks with concerns arising from operation of vessels. Oil companies use high-powered motorboats to transport personnel and enforce security. These boats travel at high speeds, create large washes, and require specific docking and mooring at jetties. During such operations, it is difficult for operators to see the fishermen, their boats or nets and the results are collisions, capsizing and loss of life. Similarly, while docking and mooring, the large size vessels pose the same risks to fishermen and fishing boats leading to boat damage, injuries, or death.

Construction activities impede livelihood activities and too, increase safety hazards. It is difficult to observe activities of artisanal fishermen from the decks of larger vessels. Secondly, the large vessels generate large bow waves and large washes that affect the safety of fishermen. Thirdly where there are established shipping channels, local communities are at risk when crossing channel. Finally, where construction, production, and general operational schedules involve lengthy waiting periods, the vessels anchor outside designated footprint, thereby considerably increasing the actual affected area. This expands the footprint and potential impact on fisheries.

Exclusion zones disrupt local transportation routes and cause substantial detours for commuters and fishermen with cost and time implications with effects of reducing returns to artisanal fishermen. The increased risks of disruption and safety concerns encourage fishermen to utilize other fishing grounds, which impacts on neighboring fishing communities and increases pressure on limited fish resources in specific areas. Such displacements involve increased distances, travel time, costs and risks. Risks are associated with risks of meeting up with storms and cases of motor breakdowns may be difficult and dangerous to solve.

Project security particularly for the enforcement of exclusion zones negatively affects communities by disrupting livelihood activities like transportation and fishing as well as threat to social stability and safety of affected communities particularly when a force is undisciplined. This can sour relationship between the sector and affected communities, affecting the sectors social license to operate.

In-migrations occurred to take advantage of employment and trade opportunities and migrants settled in the communities with new cultures sometime to the detriment of local cultures. In fishing communities, migrants often participate in fishing activities, increasing competition for scarce

resources. Furthermore, migrants who come from more developed areas and have connections with traders may have greater access to equipment, inputs, credit, and markets, with capacity of larger catches, greater returns, and ultimately market control. This competition subjects' local communities to inferior statuses and gives rise to community tensions and crime.

The interaction of construction, operational, and shipping personnel with affected communities involves the sale of food, alcohol, and drugs and the development of gambling and prostitution. Consequences are social tensions, unplanned pregnancies, sexually transmitted diseases, and alcohol and drug abuse. These issues escalate the security threats and may lead to vandalism and sabotage of oil sector installations, production and marketing. Also noted is behavior change in terms of increased opportunism and risk-taking behavior.

Movement of fishing activities from one fishing ground to another is tantamount to the movement of fishing activities to other fishing grounds, increasing the number of people affected by project development and also placing increased pressure on fisheries that may already be overexploited. This also distorts community networks and ways of life.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter follows findings that community livelihoods and the fishing business in Kaiso has been contravened and makes recommendations towards the promotion of incomes, health, education and other social services.

5.2 Incomes, Health, Education and other Social Services

5.2.1 Habitat Protection

In order to avoid habitat destruction, oil and gas prospecting should in the future, identify rich breeding sites and opt for alternative sites because access to oil wells has larger square area than breeding sites. The damage done to habitats should be repaired by way of conserving the survival aquatic life as well as reviving it as much as possible. Construction should be kept within well-marked boundaries. Pilings should be used instead of fill for in-water structures and engineers should minimize diameter and spacing. Construction systems that minimize the area of disturbance should be opted for. Impacted habitats should be restored through monitoring and capacity building of stakeholders acting in resource management. This can involve provision of education to local fisheries managers so that they can reduce risks among which is that of overfishing.

5.2.2 Eco-System Disturbance

Ecosystem disturbance can be managed through time dredging and other marine disturbance can be avoided at critical periods of spawning and migration cycles of locally important fish through managing time of works to avoid or minimize disruption to fisheries resources. To manage noise vibration, noise-dampening methods can be used through the use of vibratory hammers for pile driving when possible, or wood blocks on percussion hammers and bubble curtains around piles to dampen the vibration. Controlling the time of blasting can also help and can be done through avoiding blasting during key local species spawning or migration seasons. It would also be useful to optimize work windows so that in-water work is not conducted during the time key local species are spawning or in migration seasons.

Excessive turbidity can be managed through implementing erosion and sedimentation control. Implementing standard measures such as silt fencing, isolation of excavation areas with sheet pile and covering of exposed soil. It is also necessary to optimize work windows through avoiding

rainy seasons for earthworks. Affected areas should be re-vegetated through implementation of a re-vegetation plan and using appropriate native vegetation.

5.2.3 Pollution Management

To prevent pollution, spills should be prevented through standard best management practices (BMPs), following the International Convention for the Prevention of Pollution from Ships (MARPOL) and other international protocols. Also, it is important to use nontoxic coatings through the use of pipelines or pilings that are approved as ones with nontoxic coating.

It would also be useful to introduce evasive species through implementing a ballast water exchange protocol by following MARPOL and other international protocols. To avoid the direct killing of fish, re-stocking of key lost species should be done through the use of fish culture i.e. hatcheries or net pen culture to supply replacement resources.

In order to manage derelict munitions explosions, it is important to neutralize munitions through surveys to identify munitions and selection of appropriate removal techniques. It is also useful to minimize direct impacts to resources through surveying areas for mammals and fish schools; using fingers to scare marine resources from the exploration areas; and timely removals to avoid spawning migrations. It is also of importance to ensure frequent sector stakeholder (State Agencies and Non-State Agencies) monitoring and consultation with local fishermen to promptly identify and respond to unforeseen impacts on fisheries.

5.2.4 Compensation, Capacity Building, Education and Restoration of livelihood frameworks

These need to be put in place through National, sector, and project-level compensation and livelihood restoration frameworks developed through stakeholder engagement processes with the affected population, as well as relevant Government authorities like MAAIF and relevant MDAs. The framework should ensure that impacts serve as a benchmark for the development of the framework and, in so doing, their development may explore the existence of national or regional benchmarks, precedence, and identify concerns about sustainability of fish resources. Through the same process, agreement should be reached on monitoring frameworks.

There is need to promote the development support of alternative livelihoods to address the negative impacts on fisheries for instance agriculture through introducing cropping, and community gardens; livestock enterprises; small-scale forestry; woodlots; micro and small enterprises; offer

vocational training; encourage part-time fishermen to focus on non-fishing livelihood activities including oil related employment and facilitating early retirement for older fishermen. It is necessary to scope existing areas where affected communities and households are already operating diversified livelihood activities and where combinations of access to resources like land, seasonal production levels, and adequate markets and mark them for pilot projects. Attention has to be given to the local economic context and ensure that sufficient demand exists for the selected lines of small-scale production. Such feasible alternatives provide stability and resilience to fishing communities.

Compensations can be in cash or kind through providing food supplements to compensate for reduced access and catch during the period of temporary disturbance i.e. dredging. It can also be in the form of equipment and training. Cool boxes, nets, lines etc. can be provided to compensate for damage to equipment. Boats and motors to address reduction in access and use, relocation to more distant fishing grounds, and adaptation to new fishing practices can be done. The distribution of fishing equipment or training to improve fishing techniques should focus on increasing the capacity of affected people to exploit fish resources.

The above should be preceded by adequate resource assessment and management systems to revamp fish stocks so that proposed activities remain sustainable in the medium and long term. If this is not the case, such interventions may lead to higher intensity of resource use, through greater catches per unit of effort and higher participation rates culminating into depletion of fish stocks.

In addition to the above, it is useful to provide cash compensations and ensure that if markets exist, they are functional and offer necessary goods and services to support sustenance of livelihood capital. Cash compensations can be given in lieu of providing food and in compensation for damage to equipment. It would also be prudent to provide construction-phase employment and for on-going monitoring of resource and project impacts.

In order to restore and develop fisheries activities and livelihoods - it would be prudent to provide fishing equipment, like boats, motors and other necessary equipment and training to improve fishing techniques. These should focus on increasing the capacity of affected people to exploit fish resources. This intervention must ensure the availability of key inputs by working with traders and other stake holders.

Capacity building should be provided in fisheries resource management, boat construction, equipment maintenance and repair, fishing techniques, postharvest processing, alternative products, and aquaculture.

More can be done through: teaching courses in fisheries science/management or providing funds for local governments to hire additional qualified staff; supplying boat construction training, materials and power equipment; providing training, tools, and parts for equipment maintenance and repair; providing training and gear for improving fishing techniques; improving fishing should focus on other fishing grounds and under-utilized species to minimize risk of overexploitation and also consider alternative products like seaweed production for human dietary sector demand and as an aquaculture fish food.

It is feasible to replace natural fresh water fish capture with aquaculture although it involves access to land, capital, capacity and resources and is a fundamentally different livelihood activity with a different contribution. Following the introduction of aquaculture, training in best management practices (BMPs) is needed as well as support infrastructure for power, water, and equipment oriented towards specific markets to improve post-harvest processing. Other efforts would be to explore development of alternative value-added products and provide training and equipment for improved safety.

It would also be useful to formalize customary fishing rights and assist communities to develop and implement fisheries management plans for sustainable use through working with government and development partners to analyze and implement co-management projects and organizational development and strengthening of groups. With improvement of the local economy comes a boost to complimentary businesses such as supply of water, electricity, ice-making equipment, fuel supply and storage.

There is need to support the creation of fishermen's associations through introducing proven models for organizing, funding, and operating fishermen's associations i.e. cooperatives, and build capacity through organizational development training. Access to credit should be through facilitating fishermen's access to credit to promote entry into and expansion of fishing and support to market development should target all value chain elements i.e. fishing, processing, packing, storage, and quality control of course depending on target markets.

Physical Relocation should ensure that relocation plans align with the requirements of IFC Performance Standard 5, Land Acquisition and Involuntary Resettlement. Promotion of Sustainable Management of Small-scale and Artisanal Fisheries should involve developing interventions that promote the sustainability of small-scale and artisanal fisheries through protection and rehabilitation of fisheries systems and resources; and Local and regional management of small-scale and artisanal fisheries like community-based management systems.

5.2.5 Stakeholder Involvement and Engagement

It is highly necessary and can be preceded by a stakeholder mapping and plan but with the key purpose of addressing issues of affected households and communities. Efforts have to ensure that all stakeholders are involved, including affected fishing households and communities, participants in the fish value chain and institutional representation, provide affected households and communities with prospecting plans as early as possible and seek feedback on potential impacts. Stakeholder involvement should provide for stakeholder observation and regular focus group meetings. This should encompass monitoring of affected people's experiences. It is also important to ensure that the frequency of stakeholder meetings is correlated with periods of greatest disturbance. This helps to capture issues comprehensively before they are forgotten or distorted.

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APPENDIX I: SAMPLE OF QUESTIONNAIRE ADMINISTERED TO RESPONDENTS

Section A: General characteristics of respondent

1. Name (Optional).....

2. Organization.

3. Questionnaire No:

4. Sex

a) Male []

b) Female []

5. Age

a) 20 – 25 []

b) 26 – 35 []

c) 40 and above []

6. Marital Status

a) Married []

b) Single []

c) Widowed []

7. Educational level

a) Primary Level Education []

b) Uganda Certificate of Education []

c) Certificate []

d) Degree []

e) Masters []

8. What is your occupation?

a) Local Government Official []

b) GoU MAAIF, NEMA and other MDAs []

c) Fish Trader []

d) Fisherman []

e) Fish Folks []

- f) Canoe Owner []
- g) Oil company Official []
- h) Naval Officers []
- i) Water Traffic Officers []

9. How long have you lived in this community?

- a) Less than one year []
- b) 1 year – 5 years []
- c) 5 years – 10 years []
- d) More than 10 years []

10. How many members are in your household?

- a) More than 10 []
- b) Between 10 – 6 []
- c) Between 6 – 3 []
- d) Less than 3 []

11. How many sources of income do you have?

- a) None []
- b) 1 source []
- c) 1- 2 sources [] Please specify.....
- d) More than 2 sources [] Please specify

12. What is the monthly income you get from fishing related activities? (in Uganda Shillings)

- a) 0 – 200 []
- b) 200 – 400 []
- c) 600 – 800 []
- d) More than 800 []

Section B: Effects of oil and gas activities on the fisheries and the communities

13. What is the effect of oil and gas prospecting to be on your livelihoods and your communities?

- a) Positive effect []
- b) Negative effect []

c) Do not know []

14. What are the likely or actual positive impacts of oil and gas activities on your livelihoods and your communities?

a) Increase in job prospects within the community []

b) Improvement in social life with influx of people []

c) Development of social amenities (schools, roads, lights, etc) []

d) Others [] Specify.....

15. What are the likely or actual negative impacts of oil and gas activities on your livelihoods and your communities?

a) Low fish catch []

b) Loss of job []

c) Low level of income []

d) Legal issues against offenders []

e) Increase in accommodation cost []

f) Unable to fish at all []

g) Bad state of road and heavy vehicles []

h) Environmental degradation /pollution []

i) Harassment by naval officials []

j) Cultural change []

16. Please tick the appropriate box below:

Relationship	Strongly Agree	Agree	Strongly Disagree	Disagree
(i)				
a) Drilling activities negatively affect Incomes of Fishing Communities				
b) Drilling activities negatively affect Health of fishing communities and their business				
c) Drilling activities negatively affect capacities and education of fishing communities and their business				
d) Drilling activities negatively affect social networks and services fishing communities				
Relationship	Strongly Agree	Agree	Strongly Disagree	Disagree
(ii)				

a) Seismic activities negatively affect Incomes of Fishing Communities				
b) Seismic activities negatively affect Health of fishing communities and their business				
c) Seismic activities negatively affect capacities and education of fishing communities and their business				
d) Seismic activities negatively affect social networks and services fishing communities				
Relationship	Strongly Agree	Agree	Strongly Disagree	Disagree
(iii)				
a) Marine Transport negatively affect Incomes of Fishing Communities				
b) Marine Transport activities affect negatively affect Health of fishing communities and their business				
c) Marine Transport negatively affect capacities and education of fishing communities and their business				
d) Marine Transport negatively affect social networks and services fishing communities				

APPENDIX II. INTERVIEW GUIDE

Section A: Organization background

- 1. Name of informant (optional).....
- 2. What is your position in the institution?.....
- 3. What is the name of your institution?.....

Section B: Likely impact of oil activities on fisheries and the mitigation measures

- 4. What are the likely impacts the expanding oil and gas activities will have on fisheries?

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- 5. What kind of measure do you think have been introduced to safeguard the fisheries (including livelihoods of the local fishers and fishing communities)?

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6. Are the policies and legal framework put in place by government to safeguard the fisheries adequate?

- a) YES []
- b) NO []
- c) Do not know []
- d) Others [] Please explain.....

7. Do you think the institutions in place by government to safeguard the fisheries sector have adequate human resources and capacity to implement the policies?

- a) YES []
- b) NO []
- c) Do not know []

8. If your answer to question 7 is NO, what should be done to enhance the human resource and capacity of these institutions?

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9. To what extent do you think the local communities and stakeholders are involved in formulation of these policies and legal frameworks?

- a) Very involved []
- b) Involved []
- c) Fairly involved []
- b) Not involved []

10. What can be done to mitigate and secure a peaceful co-existence between the petroleum and the fisheries sectors?

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11. What percentage of Uganda’s oil revenue do you think should be allocated to developmental projects in oil bearing fishing communities?

- a) 1 – 2 % []
- b) 3 – 5 % []
- c) 6 – 7 % []
- d) 8 – 10 % []
- e) 11 – 14 % []
- f) Above 15% []

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APPENDIX III: CORRELATION ANALYSIS TABLES

		Correlations				
		Drilling	Income	Health	Community Capacities	Social Networks and Services
Drilling	Pearson Correlation	1	1.000**	1.000**	1.000**	1.000**
	Sig. (2-tailed)		.000	.000	.000	.000
	N	115	115	115	115	115
Income	Pearson Correlation	1.000**	1	1.000**	1.000**	1.000**
	Sig. (2-tailed)	.000		.000	.000	.000
	N	115	115	115	115	115
Health	Pearson Correlation	1.000**	1.000**	1	1.000**	1.000**
	Sig. (2-tailed)	.000	.000		.000	.000
	N	115	115	115	115	115
Community Capacities	Pearson Correlation	1.000**	1.000**	1.000**	1	1.000**
	Sig. (2-tailed)	.000	.000	.000		.000
	N	115	115	115	115	115
Social Networks and Services	Pearson Correlation	1.000**	1.000**	1.000**	1.000**	1
	Sig. (2-tailed)	.000	.000	.000	.000	
	N	115	115	115	115	115

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations

		Seismic Operations	Income	Health	Community Capacities	Social Networks and Services
Seismic Operations	Pearson Correlation	1	1.000**	1.000**	1.000**	1.000**
	Sig. (2-tailed)		.000	.000	.000	.000
	N	115	115	115	115	115
Income	Pearson Correlation	1.000**	1	1.000**	1.000**	1.000**
	Sig. (2-tailed)	.000		.000	.000	.000
	N	115	115	115	115	115
Health	Pearson Correlation	1.000**	1.000**	1	1.000**	1.000**
	Sig. (2-tailed)	.000	.000		.000	.000
	N	115	115	115	115	115
Community Capacities	Pearson Correlation	1.000**	1.000**	1.000**	1	.977
	Sig. (2-tailed)	.000	.000	.000		.000
	N	115	115	115	115	115
Social Networks and Services	Pearson Correlation	1.000**	1.000**	1.000**	1.000**	1
	Sig. (2-tailed)	.000	.000	.000	.000	
	N	115	115	115	115	115

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations

		Marine Transportation of Equipment	Income	Health	Community Capacities	Social Networks and Services
Marine Transportation of Equipment	Pearson Correlation	1	1.000**	1.000**	1.000**	1.000**
	Sig. (2-tailed)		.000	.000	.000	.000
	N	115	115	115	115	115
Income	Pearson Correlation	1.000**	1	1.000**	1.000**	1.000**
	Sig. (2-tailed)	.000		.000	.000	.000
	N	115	115	115	115	115
Health	Pearson Correlation	1.000**	1.000**	1	1.000**	1.000**
	Sig. (2-tailed)	.000	.000		.000	.000
	N	115	115	115	115	115
Community Capacities	Pearson Correlation	1.000**	1.000**	1.000**	1	1.000**
	Sig. (2-tailed)	.000	.000	.000		.000
	N	115	115	115	115	115
Social Networks and Services	Pearson Correlation	1.000**	1.000**	1.000**	1.000**	1
	Sig. (2-tailed)	.000	.000	.000	.000	
	N	115	115	115	115	115

** . Correlation is significant at the 0.01 level (2-tailed).

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